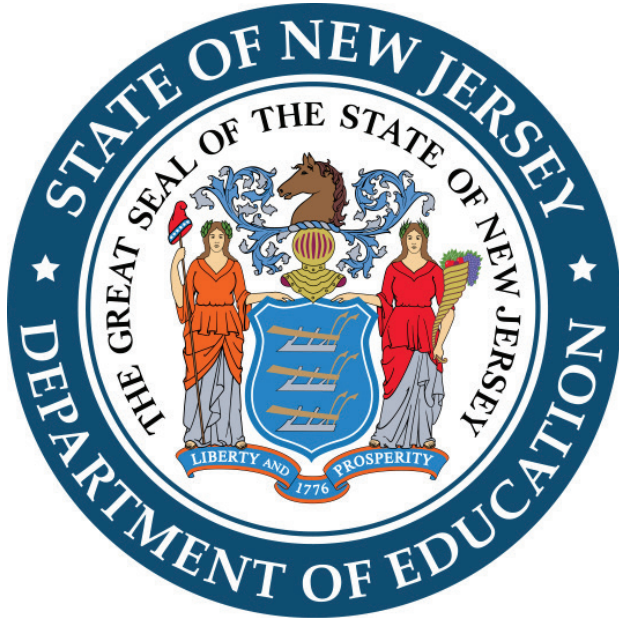




New Jersey
Student Learning
Assessments-Adaptive

Practice Test

Mathematics

Grade 7

Student Name _____



Directions:

Today, you will take the Grade 7 Mathematics Practice Test. You will be able to use a calculator. If you do not know the answer to a question, you may go on to the next question. When you finish, you may review your answers and any questions you did not answer.

1. For questions with bubbled responses, fill in the circle next to your answer choice. If you change your answer, make sure you erase your old answer completely. Do not cross out or make any marks on the other choices.
2. For questions with response boxes, write your answer neatly, clearly, and in the space provided.
3. For questions with response grids, write your answer in the answer boxes at the top of the grid.
 - Write only one digit or symbol in each answer box.
 - Use only the options listed below each box.
 - Be sure to write a negative sign or decimal point in the answer box if it is part of the answer.

4	3							} Answer boxes } Negative sign } Decimal point } Number bubbles
-	-	-	-	-	-	-	-	
.	.	.	.	.	.	.	.	
0	0	0	0	0	0	0	0	
1	1	1	1	1	1	1	1	}
2	2	2	2	2	2	2	2	
3	●	3	3	3	3	3	3	
●	4	4	4	4	4	4	4	
5	5	5	5	5	5	5	5	
6	6	6	6	6	6	6	6	
7	7	7	7	7	7	7	7	
8	8	8	8	8	8	8	8	
9	9	9	9	9	9	9	9	



1. A fraction is shown.

$$\frac{5}{9}$$

Which number is equivalent to the fraction?

- (A) 0.5
 - (B) $0.\overline{5}$
 - (C) $0.\overline{59}$
 - (D) 5.9
2. Which expression is equivalent to $\frac{-3}{4} \cdot \frac{7}{-2} \div \frac{3}{-8}$?

- (A) $\frac{-1}{2} \cdot \frac{7}{-2}$
- (B) $\frac{1}{2} \cdot \frac{7}{-2}$
- (C) $\frac{2}{1} \cdot \frac{7}{-2}$
- (D) $\frac{2}{1} \cdot \frac{7}{2}$



3. The total change in the water level of a lake from the beginning of April to the end of August was -4.7 inches. The table shows the changes in the water level from the beginning to the end of several months.

Complete the table to show possible changes in the water level, in inches, for July and August.

Water Level

Month	Change (inches)
April	3.5
May	1.1
June	-4.3
July	
August	
Total Change	-4.7

4. Which expression is equivalent to $\frac{3}{5}(5n - 12) - \frac{1}{4}(3n + 16)$?

- (A) $2\frac{1}{4}n - 3\frac{1}{5}$
- (B) $2\frac{1}{4}n - 11\frac{1}{5}$
- (C) $3\frac{3}{4}n - 3\frac{1}{5}$
- (D) $3\frac{3}{4}n + 11\frac{1}{5}$



5. Jackson buys a new shirt. The shirt is 25% off of the original price. Jackson pays a total of \$11.34 for the shirt, which includes a sales tax of 8%.

What is the original price, in dollars, of the shirt?

−	−	−	−	−	−	−	−
•	•	•	•	•	•	•	•
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9



Mathematics

6. Charmaine observes the number of times the 5:00 p.m. train is early, late, and on time. She records her observations of the train for 80 days in a row in the table shown.

Early	23 times
Late	9 times
On Time	48 times

Based on Charmaine's data, how many trains are likely to be late over the next 400 days?

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9



7. One apple costs \$0.55 at the grocery store. Customers receive one free apple for every 8 apples that they buy. Anna paid a total of \$8.80 for her apples.

How many free apples did Anna receive?

−	−	−	−	−	−	−	−
•	•	•	•	•	•	•	•
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9



8. A dance studio offers ballet and hip-hop classes. The studio has 8 ballet classes for every 12 hip-hop classes.

How many hip-hop classes are there for each ballet class?

−	−	−	−	−	−	−
•	•	•	•	•	•	•
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

9. Rafi has \$48.00 to spend on games. Each game costs \$3.00. He writes the expression shown to represent the amount of money he has after purchasing g games.

$$48 - 3g$$

He writes an equivalent expression as shown.

$$3(16 - g)$$

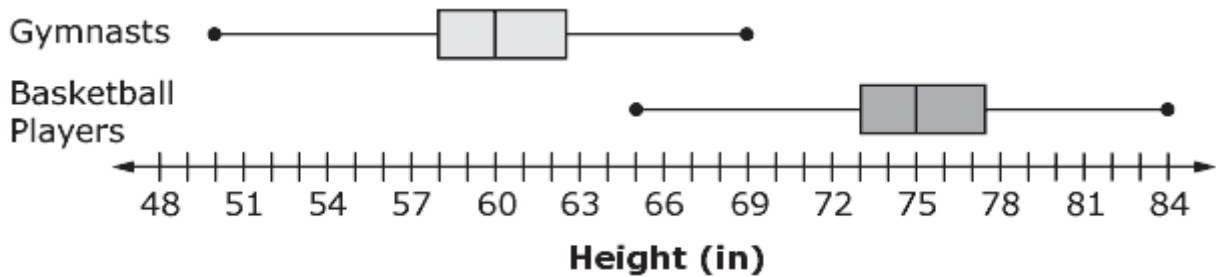
What does the number 16 represent in the equivalent expression?

- (A) the total number of games Rafi can buy
- (B) $\frac{1}{3}$ of the number of games Rafi can buy
- (C) the amount of money Rafi has after buying g games
- (D) the amount of money Rafi has after buying 3 games



10. The box plot shows a comparison of the heights of gymnasts and basketball players. The mean absolute deviation for each group is approximately 2.

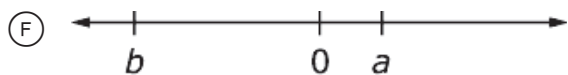
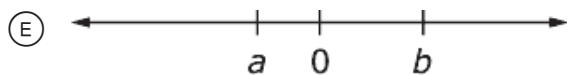
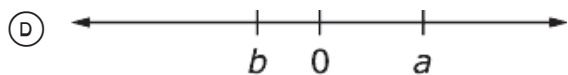
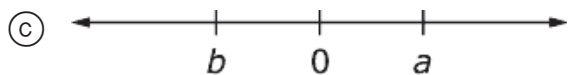
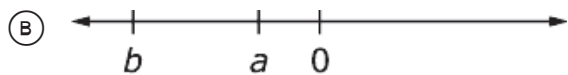
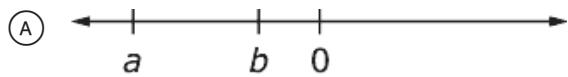
Heights of Gymnasts and Basketball Players



How many mean absolute deviations describe the difference between the two medians?



- 11.** Gary is thinking of three numbers, a , b , and c , where $b - a = c$ and $c < 0$.
Select all of the number lines that could represent Gary's numbers.



- What is Dana's walking rate in miles per hour?

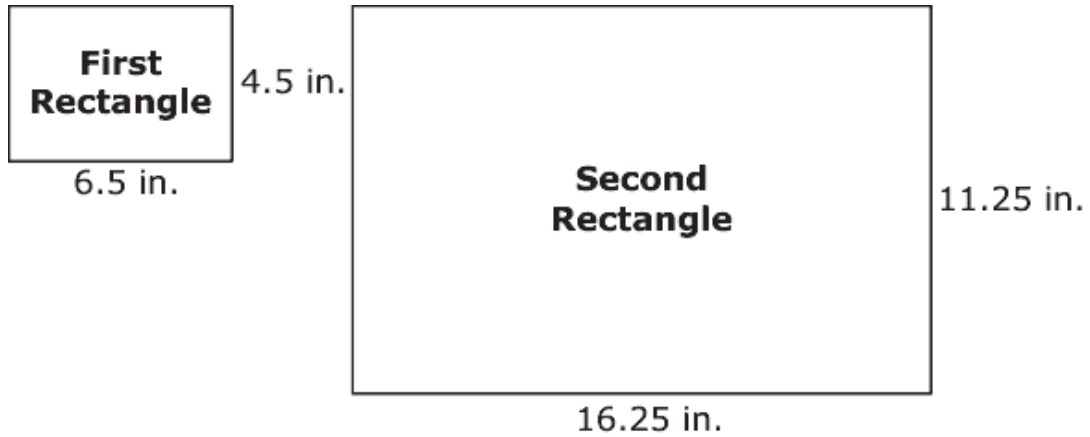
−	−	−	−	−	−	−
•	•	•	•	•	•	•
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

- What is the probability of rolling a blue or red side?

--



14. Angelo drew the two rectangles shown. The second is a scale drawing of the first.



What scale factor did Angelo use to draw the second rectangle?

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9



15. The tables show the cost for different numbers of song downloads.

Which table shows a proportional relationship?

(A)

Number of Songs	Cost
2	\$ 3.50
4	\$ 7.00
5	\$ 8.75
7	\$13.25

(C)

Number of Songs	Cost
2	\$ 3.50
4	\$ 8.00
5	\$ 8.75
7	\$13.25

(B)

Number of Songs	Cost
2	\$ 3.50
4	\$ 8.00
5	\$ 8.75
7	\$12.25

(D)

Number of Songs	Cost
2	\$ 3.50
4	\$ 7.00
5	\$ 8.75
7	\$12.25



- 16.** Raquel is building a shelf for her wall. The length of the wall is 11 feet 3 inches. The shelf will be built on the center of the wall, and the length of the shelf will be $\frac{1}{3}$ the length of the wall.

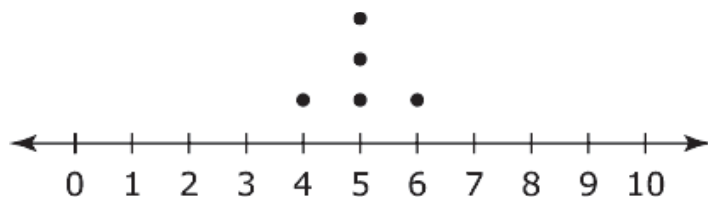
What is the distance, in inches, from the edge of the wall to the edge of the shelf?

—	—	—	—	—	—	—
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

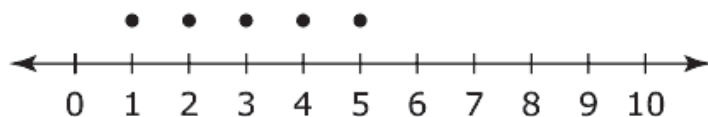


17. Which pair of data sets provides no evidence that the values of Set A are greater than the values of Set B?

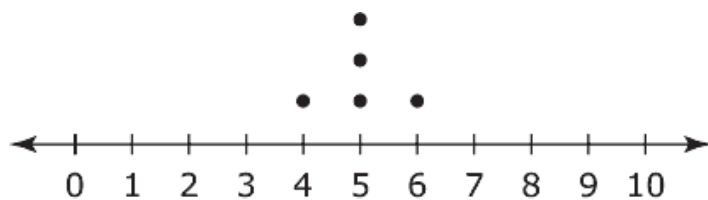
(A) Set A:



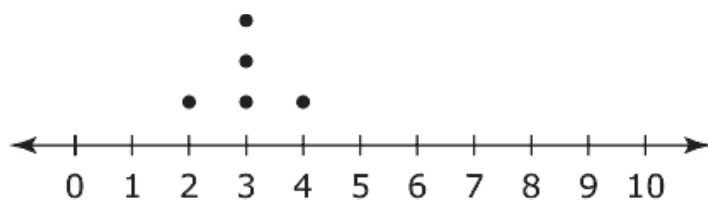
Set B:



(B) Set A:



Set B:

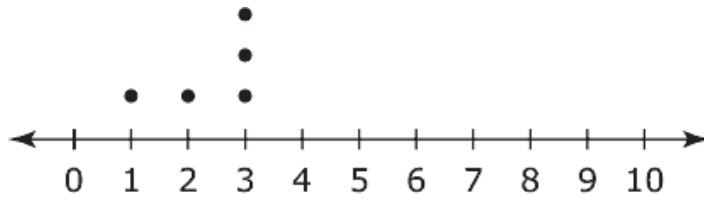




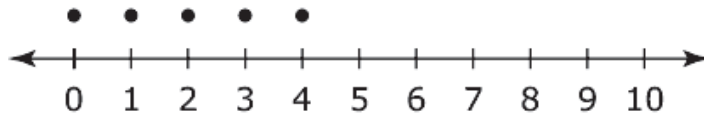
Ⓒ Set A:



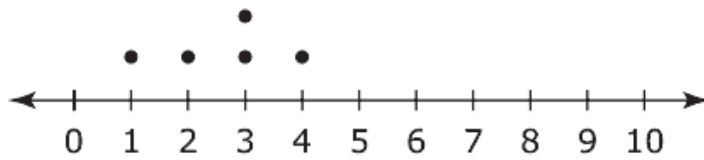
Set B:



Ⓓ Set A:



Set B:





- 18.** The circumference of a circle is 18.84 inches.

What is the diameter of the circle, to the nearest inch?

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

- 19.** A box is filled with different-colored markers. When a marker is chosen randomly from the box, it is likely that the marker will be blue.

Select all of the possible probabilities that a randomly selected marker will be blue.

- (A) 0
- (B) $\frac{1}{3}$
- (C) $\frac{2}{3}$
- (D) $\frac{3}{4}$
- (E) 1
- (F) $\frac{3}{2}$



- 20.** A furniture factory makes 250 tables each day.

Create an equation that can be used to find the total number of tables made, y , after x days.

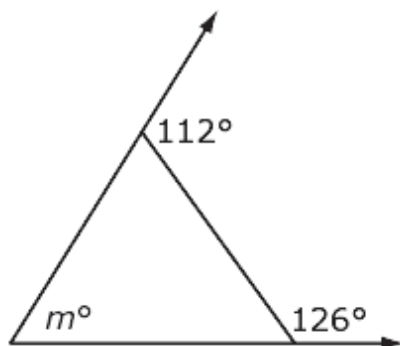
- 21.** A bowling alley charges x dollars per guest and a fixed \$50 rental fee for parties.

Which equation represents the total cost, y , for 9 guests?

- Ⓐ $y = 9x$
- Ⓑ $y = 9x + 41$
- Ⓒ $y = 9x + 50$
- Ⓓ $y = 50x + 9$



22. A figure is shown.



What is the value of m ?

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9



Paper Practice Test Answer and Alignment Document: Math

Grade 7 Mathematics

Item Sequence	Answer Key	Standards Alignment
1	B	7.NS.A.2d
2	C	7.NS.A.2c
3	any two numbers whose sum is -5	7.NS.A.3
4	B	7.EE.A.1
5	14, or any equivalent value	7.RP.A.3
6	45, or any equivalent value	7.SP.C.6
7	2, or any equivalent value	7.EE.B.4a
8	1.5, or any equivalent value	7.RP.A.2b
9	A	7.EE.A.2
10	7.5, or any equivalent value	7.SP.B.3
11	B, C, D, F	7.NS.A.1c
12	3, or any equivalent value	7.RP.A.1
13	$\frac{4}{6}$, or any equivalent value	7.SP.C.7a
14	2.5, or any equivalent value	7.G.A.1
15	D	7.RP.A.2a
16	45, or any equivalent value	7.EE.B.3
17	D	7.SP.B.4
18	any value between 5.99 and 6	7.G.B.4
19	C, D	7.SP.C.5
20	$y = 250x$, or any equivalent equation	7.RP.A.2c
21	C	7.EE.B.4a
22	58, or any equivalent value	7.G.B.5